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Engineering Innovation Challenge 2022

Med IT Ez

Team U-2

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Project Abstract

The increasing population of elderly in Singapore calls for a change in our healthcare system to better accommodate the older generation. However, the healthcare sector faces challenges in recruiting health workers to meet the manpower demand. Therefore, innovative solutions such as autonomous machines should be considered to cater for better healthcare for the elderly. This project aims to solve the challenges faced by elderlies such as immobility and forgetfulness, providing a solution for the social isolation of the elderlies. The solution by Team U-2, Med IT Ez, an autonomous machine that acts as a cabinet has been proposed. The purpose of the machine is to keep medications secure, detect abnormal activities, and remind the user to take the medication with the help of cloud-based data. It records information about the stored medicine, as well as the time that the user has to take medication to remind the user of what medication to take and when to take it. Thus, avoiding overdosage or the event of forgetting to take medication. The machine is also equipped with a motion-tracking camera that is empowered by AI, to detect and analyse any abnormal activity that may contain potential health risks to the user's health status. The machine works together with an app, as a means of the input and output of information in and out of the cloud-based data. With this application of AI, a more inclusive and sustainable society that meets the current needs of elderlies will be achieved.

Project Report

1. Background

With increasing life expectancy and low fertility rates, 25 percent of the population in Singapore is predicted to reach the age of 65 and older by 2030, dominating a quarter of the whole population of Singapore. (Lin, 2021) This rapid shifting in demographics would require the healthcare system to be prepared to accommodate the daily needs of our older generation. In today's world, the older generation faces challenges and problems such as immobility and forgetfulness which will definitely increase the difficulty of completing everyday tasks.

Based on the Singapore Census of Population, the number of residents in resident households aged 65 years and over who had mobility issues grew from around 25,500 to 50,000 in a mere twenty years, i.e. from 2000 to 2020. (Ministry of Health, Singapore, 2021) Diseases like Alzheimer's and dementia also cause elderlies to forget daily life routines, such as taking medicine and buying groceries. According to research from the Ministry of Health, Singapore, 1 in 10 seniors who are aged 60 and above in Singapore face challenges such as dementia. This number is expected to rise in the future with increased life expectancy.

On top of these complications, most elderlies often experience neglect, sometimes even by their own children, as the children have their own obligations and responsibilities too. Sadly, although life expectancy has increased, the quality of life seems to have stagnated throughout the past few years.

Therefore, new solutions should be developed to help elderlies lead an independent and healthy life during old age. Increasing the number of healthcare workers or caregivers seems like a logical remedy. However, the healthcare sector faces severe challenges in recruiting enough workers to fulfil the manpower needs; other more innovative ways of catering to elderlies' healthcare needs, such as autonomous machines, should be considered for more holistic healthcare for the elderly.

2. Problem statement

How can AI be implemented to overcome the challenges faced by elderlies such as immobility and forgetfulness to create a more inclusive and sustainable society?

3. Machine Design

Material: Plastic, LED, Buzzer, Camera, Electric lock, Button

Dimension: 35cm (L) x 25cm (W) x 30cm (H)

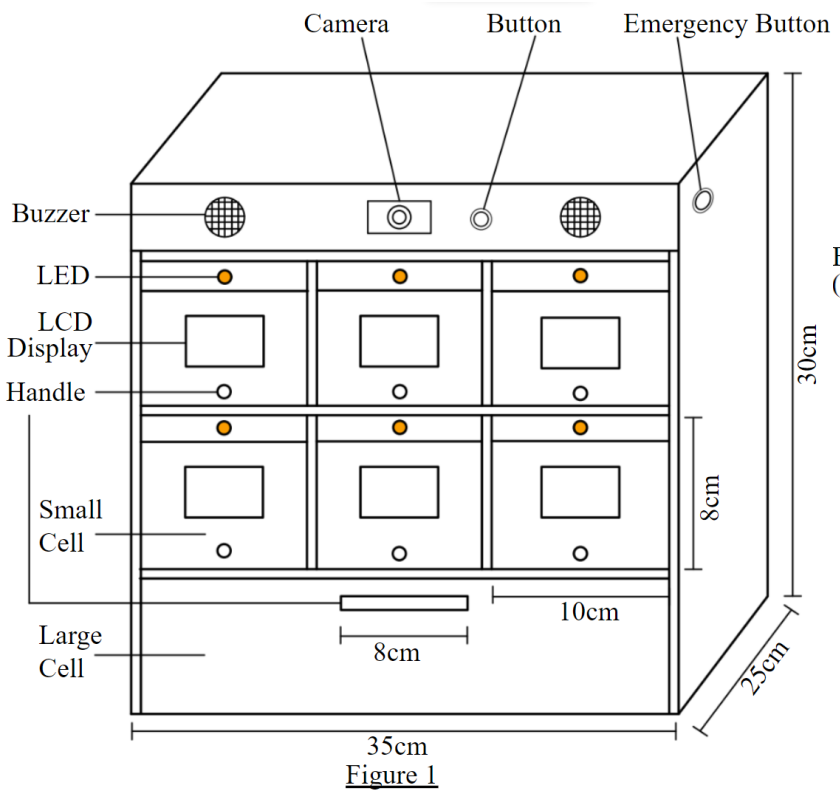
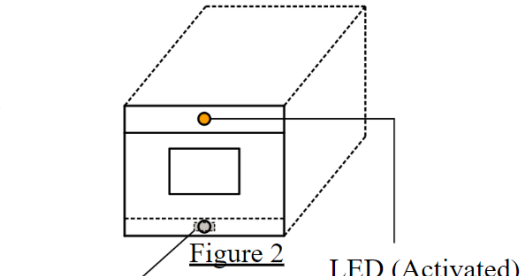


Figure 1


Figure 2

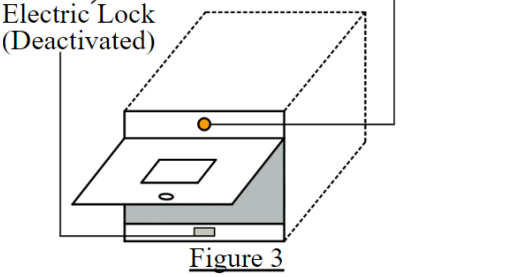


Figure 3

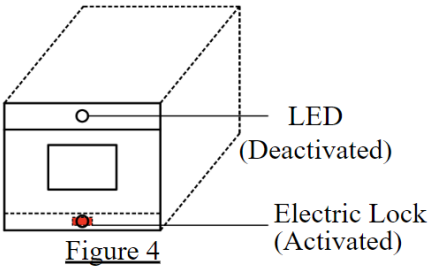


Figure 4

4. Features of components (Figure 1)

1. **Buzzers:** Making different sounds, ringing sound as an alarm to remind the elderly to take medicines, beeping sound to remind them to put back the medicine and emergency siren sound for emergency cases (eg. calling for help from nearby).
2. **Camera:** Detect and track the activities of the elderly & allow guardians to monitor the elderlies through the app.
3. **Cells:** Small: store medicine (up to 6 different types); Large: store medicine (as backup/for refill)
4. **Button:** Press to switch off the buzzer after taking medication.
5. **Emergency Button:** Emergency sirens will be activated to alert passers-by that the user is in danger.
6. **LED:** LEDs that correspond to the medicines to be taken will light up.
7. **Electric Lock:** Avoid the users from taking the wrong medicine by locking up the cell of the medicine that should not be taken, it also includes a weight sensor that will trigger the lock only after the medicine is returned.
8. **LCD Display:** Show the medicine dosage that the user needs to take for the respective type of medicine or any other significant information.

5. Machine Mechanism

1. The machine will retrieve user data from the cloud database.
2. The buzzer of the machine will signal the user with sounds when it is time to take medicine
3. When the cell is opened for a long duration or when the weight sensor senses that the medicine is not put back, it can be turned off manually with the app or button.
4. The electric lock of corresponding cells will deactivate and the LED will light up to indicate which cells should the user take their medicine from. (Figure 2)
5. Users will need to push the button to mute the buzzer after taking their medication.
6. For the other cells and cells in which the medicine has been returned (Figure 3), the LED will turn off, and the mini electric lock will activate the electromagnet to avoid the users from overdosing or taking them at the wrong time (Figure 4).
7. The smart camera does real-time motion tracking to detect abnormalities (e.g. falling down).
8. The user can modify any cells to store medicine that needs to be taken at any time for emergency use (e.g. ibuprofen, nitroglycerin) by using the app, these cells will not be locked by default.
9. If multiple types of medicine are needed to be taken at a time, the next cell will only be unlocked after the first medicine is returned back into its cell.

6. Use of Mobile Application

1. The downloaded app is for users to key in their personal data and the data of medicine such as the expiry date, quantities, and types of medicine.
2. The app can be used to modify the cells' electric locks. For example, to store emergency medicine that might need to be consumed at any time. (Emergency medicine: fast-acting medicine that ought to be taken by the users when facing life-threatening situations which can help alleviate the condition in a short period of time.)
3. Use the app to control the LCD display label by entering the content into the app.
4. When the machine starts beeping, the app will simultaneously send reminders to inform the users to take their medicine.
5. The app will update a calendar showing when the users have or have not taken their medication.
6. The app can alert or remind the user's caregivers or family members whenever an abnormal activity has occurred, along with a statistical report.
7. Emergency calls to the user's caregivers or family members will happen whenever the user manually activates it via the app. It can be cancelled when users close it with the app to avoid misunderstandings such as users mistakenly calling it or accidentally pressing the emergency button.
8. Users' caregivers and family members can view the real-time video by controlling the built-in camera with the app to check the user's condition at any time.

7. Design of cloud-based management system

1. The cloud database stores medicine information and other activities, which allows the system to start scheduling the consumption timings, analysing storage data and monitoring the machine.
2. Data recorded by the smart camera such as the average duration and timing of users appearing in the camera, the numbers of users passing by the camera, the speed of users passing by the camera, the number of user's attempts to open the medicine cells, etc. will be sent to the cloud database for further calculation of user's health status.
3. Abnormal data such as always forgetting to take medicine, not putting medicine packages back in the cell, forgetting to refill etc. will be highlighted.
4. The system will do further responses for highlighted abnormal data such as alerting the user's caregivers and family members via the app for them to take note of the user's health status.
5. The types and quantity of medicine refilled and taken are also recorded as data. Therefore, whenever any type of medicine is running out, the system will send a message to the user or the user's caregivers or family members to remind them that the medicines are running out.

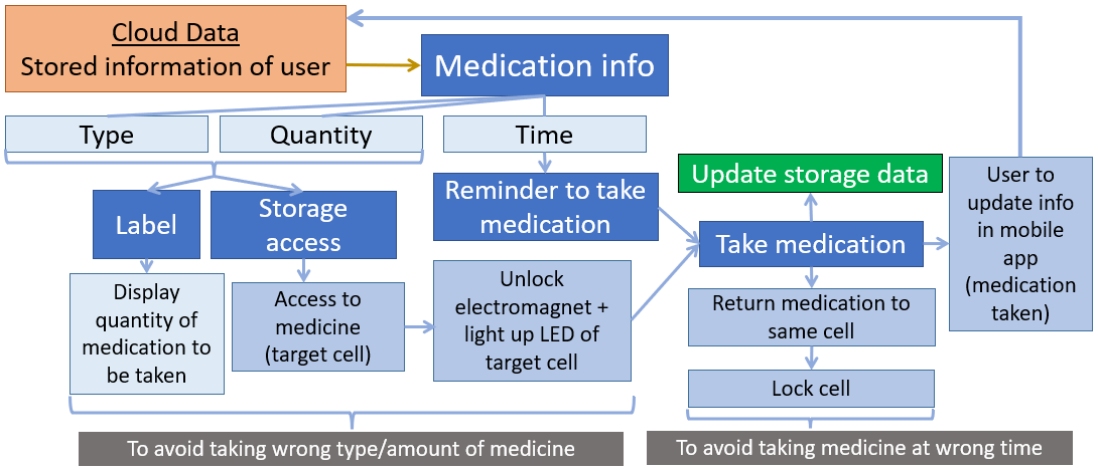


Figure 5. Basic framework of machine

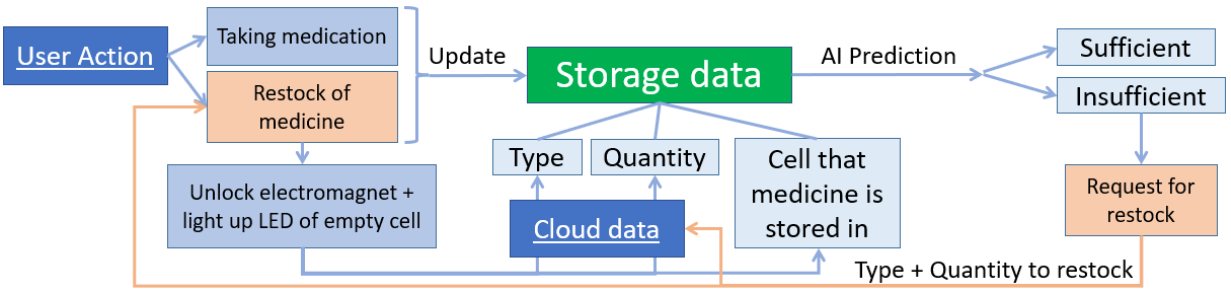


Figure 6. Basic framework of Storage system

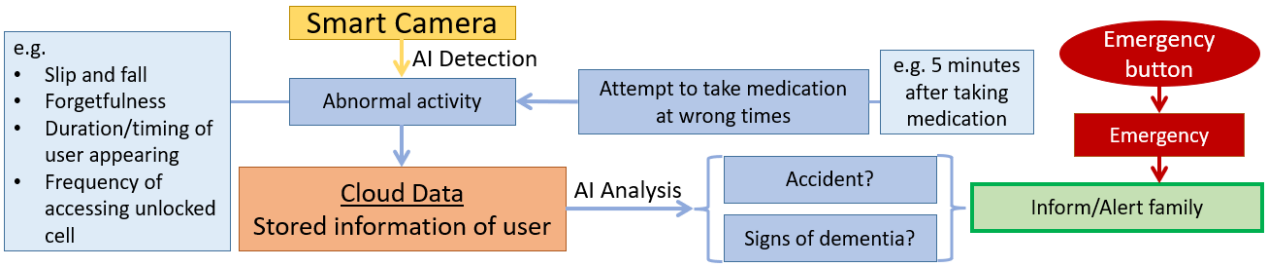


Figure 7. Basic framework of abnormal data detection

8. Cost

Resources	Quantity	Cost per item (SGD)	Cost per unit (SGD)
Customised plastic box with drawers	1	80	80
Motion tracking camera	1	100	100
LED	6	5	30
Buzzer	2	5	10
Button	2	5	10
Mini electric lock with electromagnetic and weight sensor	7	40	280
Wiring			60
Raspberry Pi Model	1	300	300
Digital LCD Display	6	15	90
		Total	***960***

Table 1

9. Risk and Safety

1. Mechanical Hazard: The doors of the cells may cause injuries if body parts are in the way when it is being closed.
   ➔ Soft pads are added to the edges of the cell to cushion body parts that are caught in the drawer.
   ➔ Soft closers are added to slow down the closing speed of the door.
2. Electrical Hazard: One might get shocked or burned when coming into contact with the live parts of the circuit.
   ➔ All wirings of the Med IT Ez machine are well insulated and hidden from the user to avoid incidents of electric shock.
   ➔ The button is covered with a flexible plastic shield to avoid shocks if users' hands are wet.
3. Health Hazard: Users' health can be at risk if the product is not used properly. Some common hazards include overdosing, underdosing and contamination.
   ➔ AI is used to calculate the instantaneous amount of drugs stored in Med IT Ez machine at any time (i.e. total input-output) so that it can track if the user is overdosing or underdosing.
   ➔ Separate compartments are used to store different types of medicine which avoids contamination between drugs.

10. Conclusion

"Med IT Ez" implements cloud application, motion detection and tracking, and smart machines to organise the routine of taking medicine, provide real-time data to third parties such as family members or caregivers of the elderly and secure the wellbeing of elderlies by avoiding social isolation. This project will help to shape a more inclusive and sustainable society by fulfilling the current needs of elderlies.

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